Fiscal Year:	FY 2012	**Task Last Updated:**	FY 11/18/2011
PI Name:	Wyrobek, Andrew Ph.D.		
Project Title:	Molecular Characterization of Long-Term Degenerative CNS Risks from Space Radiation		
Division Name:	Human Research		
Program/Discipline:	HUMAN RESEARCH		
Program/Discipline--Element/Subdiscipline:	HUMAN RESEARCH--Radiation health		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	(1) **SR**:Space Radiation		
Human Research Program Risks:	(1) **BMed**:Risk of Adverse Cognitive or Behavioral Conditions and Psychiatric Disorders		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Comments:	For immediate assistance please contact my administrator Caron LaMarsh clamarsh@lbl.gov, 510.486.5317		
Project Type:	Ground	**Solicitation / Funding Source:**	2010 Space Radiobiology NNJ10ZSA001N
Start Date:	01/01/2011	**End Date:**	12/31/2013
No. of Post Docs:		**No. of PhD Degrees:**	0
No. of PhD Candidates:		**No. of Master' Degrees:**	0
No. of Master's Candidates:		**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	1	**Monitoring Center:**	NASA ARC
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Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:	Francesco Marchetti left LBNL and took a position in Health Canada. Trent Northen, Xiu Lowe, Antoine Snijders, and Andrew Wyrobek have divided his workload.		
COI Name (Institution):	Snijders, Antoine (Lawrence Berkeley National Laboratory) Lowe, Xiu (Lawrence Berkeley National Laboratory) Northen, Trent R (Life Sciences Division, Lawrence Berkeley National Laboratory, Berkeley)		
Grant/Contract No.:	NNA11AA58I		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	The objectives of the proposed research are to identify and characterize the signaling and cellular defects associated with persistent long-term CNS deficits in neurogenesis, memory and behavior after whole body exposures to low-fluence HZE nuclei and proton space radiation exposures using the rat model. This project builds on our preliminary studies of persistent CNS tissue damage and signaling defects in the hippocampus and choroid plexus (CP) of 56Fe and 12C exposed animals. We are investigating the dose and time response of persistent signaling and cellular defects within the CP and neurogenesis zones of the hippocampus, and the composition of the cerebrospinal fluid (CSF). Our hypothesis is that radiation damage to the CP can persist for the life of the individual, producing abnormal CSF that lacks critical metabolites and neurogenic factors needed for a normal CNS microenvironment, neurogenesis and cognitive functions. The project was revised to be 3 years in duration (2011-2013) with the following revised aims. Aim 1: Characterize the dose dependence of persistent signaling and cellular defects in the CP and hippocampal zones of neurogenesis of 12C and 56Fe-exposed rats that have persistent deficits in their cognitive function and motor behavior. Aim 2: Characterize the time course of cellular and signaling defects in the CP of rats irradiated with 56Fe beam and determine the temporal relationship between CP defects and abnormal hippocampal neurogenesis. Aim 3: Determine whether irradiated animals with defective CP also produce CSF deficient in critical metabolites or proteins that are important for adult neurogenesis, using proteomic, metabolomic, and neurosphere cultures.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	This research will lay a new foundation for understanding the role of the choroid plexus in maintaining normal cerebral spinal fluid (CSF) and the central nervous system microenvironment for normal neurogenesis and normal cognitive functions in individuals exposed to space radiation. This research will also impact our understanding of how CSF changes in normal aging and its association with age-related changes in cognitive functions.
Task Progress:	Neurocognitive deficits and choroid plexus (CP) defects persisted to old age after cranial irradiation of young adult rats with 56Fe and 12C beams. We hypothesized that exposure to HZE radiation (56Fe and 12C) of rats at young ages induces persistent neurocognitive deficits that are due, in part, to radiation damage of the central nervous system (CNS) microenvironment. We found severe damage of the epithelial architecture of the CP at very old ages, especially after 1Gy 56Fe exposures. The modulated transcript signature of 56Fe irradiated brain tissue correlated with the degree of cognitive impairment in the same animals, and involved a significant Huntington's disease subnetwork. The expression of certain genes showed down-regulation at ~ 1.5 years post exposures in CP of 56Fe and 12C irradiated rats. These findings identify the CP, which produces CSF and maintains CNS homeostasis, as a critical target for persistent radiation damage to the CNS, affecting both memory and behavior. Choroid plexus defects, CSF defects, and abnormal hippocampal neurogenesis in rats irradiated with 56Fe. We recently completed an exposure study with 56Fe (whole body exposures; sham, 0.1 and 1.0Gy; using retired breeder males; plus ~10 young adult males, as reference). Animals were sampled for CSF at 2 month post-exposure, and the following samples at 3 month (blood plasma, hypocampus and cortex tissue, and one hemisphere for tissue sectioning and fluorescence immunohistochemisty). These samples are being used to investigate the time course of radiation-induced CNS damage, specifically in the choroid plexus and cerebral spinal fluid. Characterization of the early CNS stress biomarkers and profiles associated with neuropsychiatric diseases. Neuropsychiatric disorders (including dementia) have high personal, family, and social costs. Although many neuropsychiatric disorders share common patterns of symptoms and treatments, there are no validated biomarkers that define the underlying molecular mechanisms in the central nervous system (CNS). We hypothesize that there are early and common molecular changes in the CNS that will serve as sensitive indicators of CNS molecular stress and that will be predictive of neuropathological changes resulted in increasing the risk for neuropsychiatric diseases. Using the rodent model, we showed that systemic exposure to three diverse CNS stressors with different mechanisms of action (ketamine, low-dose and high-dose ionizing radiation, interferon-a) induced the expression of troponin T1 (Tnnt 1) within hours in adult mouse brain tissue. Tnnt 1 expression was induced in neuronal (not glial) cells, the hippocampal zone of neurogenesis, cerebral cortex, amygdale, and choroid plexus, which are important CNS locations in behavior and mental health. We also identified nine neural signaling pathways that showed a high degree of concordance in their transcriptional response in mouse brain tissue four hours after low-dose irradiation, in the aging human brain (unirradiated), and in brain tissue from patients with Alzheimer's disease. Our studies provide new molecular information on shared mechanisms and expression profiles of diverse neuropsychiatric disorders. This knowledge will be fundamental for developing molecular signatures of early CNS stress biomarker for early diagnosis and treatment of neuropsychiatric diseases.
Bibliography Type:	Description: (Last Updated: 06/19/2025)
Abstracts for Journals and Proceedings	Lowe XR, Rabin BM, Marchetti F, Bhatnager S, Snijders A, Wyrobek AJ. "Choroid plexus: a critical target of persistent CNS damage after space radiation with 56Fe and 12C." Presented at the 22nd Annual Space Radiation Investigators' Workshop, League City, TX, September 18-21, 2011. 22nd Annual Space Radiation Investigators' Workshop, League City, TX, September 18-21, 2011. http://www.dsls.usra.edu/meetings/radiation2011/pdf/7107.pdf , Sep-2011
Articles in Peer-reviewed Journals	Lowe XR, Wyrobek AJ. "Characterization of the early CNS stress biomarkers and profiles associated with neuropsychiatric diseases." Current Genomics. In press, December 2011. , Dec-2011